

Abdellatif Salah Eldin

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Education

B.Sc. in Computer Science (AI Track) — Arab Academy for Science, Technology and Maritime Transport (AASTMT), Egypt 2022 – 2026

- GPA: 3.56/4.0, Excellent with Honors
- Selected grades: Mathematical Foundations for AI (A+), Advanced Statistics (A+), Explainable AI (A+), Computer Vision (A), Deep Learning (A-), Natural Language Processing (A-), Project I and II (A+, A+)

Research

Medical Image Segmentation: HTAN (Hyper TransAttUNet) — AASTMT 2025 – 2026
Supervisor: Assoc. Prof. Radwa Fathalla, College of Computing and Information Technology

- Co-developed HTAN, a segmentation architecture that adds manifold-constrained hyper-connections to the TransAttUNet bottleneck. On ISIC-2018 (skin lesions), GlaS (glands), and Data Science Bowl 2018 (nuclei), it improved Dice by 1.05, 3.57, and 1.07 points over the reproduced baseline.
- Responsible for data preprocessing, model architecture design, training optimization, and evaluation, including ablations on the number of residual streams and mHC blocks.
- Fixed NaN training failures caused by fp16 overflow in the Sinkhorn-Knopp projection by computing it in FP32 and adding gradient clipping.
- Co-authoring the paper, in preparation for submission to a peer-reviewed AI or medical imaging journal.

Biomedical Retrieval-Augmented Generation for Clinical Question Answering — AASTMT 2025 – 2026
Supervisor: Assoc. Prof. Radwa Fathalla, College of Computing and Information Technology

- Designed and built a retrieval-augmented question-answering system over PubMed, PMC, and clinical guideline documents for three disease areas (melanoma, skin cancer, multiple myeloma), using dense-sparse hybrid retrieval and cross-encoder reranking to improve passage relevance.
- Added query rewriting, out-of-scope detection, and citation-grounded generation so the system declines to answer rather than producing an unsupported response.
- Evaluated the system on -questions benchmark across the three disease areas, scoring approximately 0.91 relevance, 0.95 coverage, and 0.95 answer faithfulness; treating this as a preliminary result given the small benchmark size.
- Currently investigating multimodal retrieval that combines clinical text with medical imaging findings.

Projects

MediLink: Multimodal Medical AI Platform 2025 – 2026

- Designed and built a full-stack medical AI application that connects my HTAN segmentation model and biomedical RAG system through an LLM-orchestrated pipeline.
- Led integration of the computer vision module (CNN and Transformer-based models for X-ray and MRI interpretation) with the NLP retrieval and generation pipeline.
- Orchestrated retrieval, vision inference, and language generation in a single agentic pipeline using LangGraph.
- Built the API gateways and service architecture connecting the vision model, RAG engine, and LLM components; containerized with Docker and deployed on AWS.

Efficient LLMs for Clinical Question Answering

2025

- Fine-tuned a lightweight language model for clinical question answering using LoRA and QLoRA, with a FAISS-based retrieval layer, and evaluated results on subsets of MedQA and PubMedQA.

Experience

Researcher, AASTMT — Alexandria, Egypt

Sep 2025 – Present

- Contributing to a research paper, “A Full-Stack Autonomous Healthcare System,” that integrates computer vision and NLP components into a unified diagnostic pipeline.

Certifications

Retrieval Augmented Generation (RAG) — DeepLearning.AI

2026

Natural Language Processing with Attention Models — DeepLearning.AI

2026

Oracle Cloud Infrastructure 2025 Certified Data Science Professional — Oracle

Sep 2025

Supervised Machine Learning; Advanced Learning Algorithms — Stanford University (Coursera)

2025

Unsupervised Learning, Recommenders, and Reinforcement Learning — DeepLearning.AI

2025

In progress: NLP with Deep Learning (CS224N); Mining Massive Datasets — Stanford University

Technical Skills

Languages: Python, SQL, Java, C++

Computer Vision: Medical Image Segmentation and Classification, CNNs, Vision Transformers, SAM, MedCLIP, Foundation Model Fine-tuning

NLP / LLMs / Retrieval: Retrieval-Augmented Generation, Hybrid Retrieval, Cross-Encoder Reranking, LLM Fine-tuning (LoRA/QLoRA), Explainable AI

Frameworks and Tools: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, sentence-transformers, Weaviate, FAISS, LangGraph, Docker, AWS, Git, Linux, CUDA, Jupyter, pandas, NumPy